

# Baoli Hao

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🎓 [Baoli's Google Scholar](#)

## EDUCATION

### Illinois Institute of Technology

*Ph.D. in Applied Mathematics*

Aug. 2022 – Expected Apr. 2027

Chicago, IL

- **Advisors:** Prof. Chun Liu & Prof. Ming Zhong | **GPA:** 3.93/4.0

### City University of Hong Kong

*BSc in Computing Mathematics (First Class Honours)*

Aug. 2017 – Apr. 2021

Hong Kong

- **GPA:** 3.7/4.3 | **Minor:** Computer Science

## RESEARCH INTERESTS

My research focuses on **inverse problems** and **scientific machine learning (SciML)** for complex dynamical systems, with an emphasis on exploiting mathematical and physical structure for stable, interpretable, and scientifically meaningful data-driven learning and scientific computation. My work also connects these problems with **multiscale modeling**, **nonlinear PDEs**, and **interacting particle systems**.

## PUBLICATIONS & PREPRINTS

1. **Baoli Hao**, Kamrun Mily, Annalisa Quaini, and Ming Zhong. A finite element framework for simulating residential burglary in realistic urban geometries. *Mathematical Models and Methods in Applied Sciences*, 36(5), 1019–1050, 2026.
2. Kin Shing Chan, **Baoli Hao**, Kei Fong Lam, and Björn Stinner. On a phase field model for binary mixtures of micropolar fluids with non-matched densities and moving contact lines. *Interfaces and Free Boundaries*, published online first, 2026.
3. **Baoli Hao**, Chun Liu, Ulisses Braga-Neto, Lifan Wang, and Ming Zhong. Stability in training PINNs for stiff PDEs: Why initial conditions matter. *Foundations of Data Science*, Early Access, 2026.
4. **Baoli Hao**, Ming Zhong, and Kevin O’Keeffe. Attractive and repulsive interactions in the one-dimensional swarmalator model. *Physical Review E*, 108(6), 064214, 2023.
5. **Baoli Hao**, Kamrun Mily, Annalisa Quaini, and Ming Zhong. Crime hotspot dynamics in residential burglary models with police response. arXiv:2605.17709, 2026.
6. **Baoli Hao**, Fanze Kong, Kei Fong Lam, and Chun Liu. Global finite-energy weak solutions and sharp entropy decay for a Poisson–Nernst–Planck system with interspecies drag and steric effects. arXiv:2607.21742, 2026.
7. **Baoli Hao**, Chenxi Hu, Ming Zhong, and Ren Wang. Event-structured physics-informed neural networks for differentiable critical clearing boundaries. arXiv:2607.27681, 2026.

## MANUSCRIPTS

1. **Baoli Hao**, Mauro Maggioni, and Ming Zhong. One shot of learning of collective dynamics from equilibrium in interacting particle systems. Manuscript in preparation, 2026.
2. **Baoli Hao**, Mauro Maggioni, and Ming Zhong. Learning interaction topology and kernels in interacting particle systems. Manuscript in preparation, 2026.
3. **Baoli Hao**, Mauro Maggioni, and Ming Zhong. Data-driven discovery of gravitational structure from solar-system ephemerides. Manuscript in preparation, 2026.

4. **Baoli Hao**, Annalisa Quaini, and Ming Zhong. Discovery of hidden crime dynamics, Part I: Parametric structures. Manuscript in preparation, 2026.
5. **Baoli Hao**, Xiaofan Li, Chun Liu, Yiwei Wang, and Ming Zhong. Data-driven discovery of local and nonlocal constitutive physics in ion transport. Manuscript in preparation, 2026.

## TALKS & PRESENTATIONS

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|-------------------------------------------------------------------------------------------------------------|---------------|
| <b>SIAM Annual Meeting</b>                                                                                  | Jul. 2026     |
| Title: <i>Modeling, Simulation and Learning for Urban Crime Dynamics</i>                                    | Cleveland, OH |
| <b>SIAM Texas-Louisiana Sectional Meeting</b>                                                               | Sept. 2025    |
| Title: <i>Stability in training PINNs for stiff PDEs: Why initial conditions matter</i>                     | Austin, TX    |
| <b>Central AMS Sectional Meeting</b>                                                                        | Apr. 2025     |
| Title: <i>Modeling and analysis of Poisson-Nernst-Planck systems with steric effects and relative drags</i> | Lawrence, KS  |

## INTERNSHIPS

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| <b>HKU Business School, The University of Hong Kong</b>                                                                                                                                                                                                                                        | Jul. 2021 – May 2022 |
| <i>Data Scientist</i>                                                                                                                                                                                                                                                                          | Hong Kong            |
| <ul style="list-style-type: none"><li>• Supervisor: Dr. Qingchen Wang</li><li>• Developed machine-learning and predictive models for risk stratification and survival analysis using large-scale electronic patient record data covering approximately 90% of Hong Kongs population.</li></ul> |                      |

## SERVICE

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|-----------------------------------------------------------------------------------------------------|------------------------|
| <i>Coordinator</i> , <u>Multiscale Modeling and Computation Seminar</u> , Illinois Tech             | Aug. 2022 – Present    |
| <i>Sponsorship and Student Support Chair</i> , <u>ACM HyperText 2025 Conference</u> , Illinois Tech | Oct. 2024 – Sept. 2025 |
| <i>Reviewer</i> , AAAI Conference on Artificial Intelligence (AAAI 2026)                            | Sept. 2025             |
| <i>Reviewer</i> , ACM Transactions on Modeling and Computer Simulation (TOMACS)                     | Jan. 2026              |
| <i>Program Committee Member</i> , AAAI Conference on Artificial Intelligence (AAAI 2027)            | Aug. 2026              |

## TEACHING

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|------------------------------------------------------------------------------------|---------------------|
| <b>Illinois Institute of Technology</b>                                            |                     |
| MATH Graduate Qualifying Exam Analysis Prep Session ( <i>Leader &amp; Mentor</i> ) | Fall 2025 – Present |
| MATH 500: Applied Analysis I ( <i>Teaching Assistant</i> )                         | Fall 2025           |
| MATH 578: Computational Mathematics II ( <i>Grader</i> )                           | Spring 2025         |
| MATH 151: Calculus I ( <i>Recitation Instructor</i> )                              | Spring 2023         |

## HONORS & AWARDS

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|-----------------------------------------------------------------------|-------------|
| <b>Menger Lecture Poster Award</b> , Illinois Institute of Technology | Apr. 2026   |
| <b>Full Tuition Scholarship</b> , City University of Hong Kong        | 2017 – 2021 |
| <b>Deans List</b> , City University of Hong Kong                      | 2018 – 2021 |

## SKILLS

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**Programming** : Python, R, MATLAB, SQL  
**Frameworks & Libraries**: PyTorch, TensorFlow, FEniCS, Scikit-learn, SciPy, NumPy, Chebfun  
**Languages** : English (Fluent), Mandarin (Native), Cantonese (Good)  
**Others** : LaTeX, Microsoft Office, Git